

JASPER, AB, Canada

WMO: 728880

Lat: 52.880N

Lon: 118.070W

Elev: 1062

StdP: 89.20

Time Zone: -7.00 (NAM)

Period: 82-92

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -31.1	(c) -27.5	(d) -33.7	(e) 0.2	(f) -30.5	(g) -30.4	(h) 0.3	(i) -27.3	(j) 7.6	(k) -13.6	(l) 6.9	(m) -12.4	(n) 2.1	(o) 230	(p) 0.623

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.7	(c) 27.4	(d) 14.4	(e) 25.4	(f) 13.7	(g) 23.6	(h) 13.0	(i) 15.3	(j) 24.7	(k) 14.5	(l) 23.3	(m) 13.7	(n) 21.7	(o) 2.4	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.3	10.2	16.3	11.3	9.5	15.3	10.3	8.9	15.1	46.5	24.8	44.0	23.4	41.5	21.7	18.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max					
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
6.7	5.9	5.2	DB	-35.1	30.5	4.7	0.9	-38.5	31.1	-41.3	31.7	-43.9	32.2	-47.4	32.9	
			WB	-34.7	17.4	4.2	0.6	-37.8	17.8	-40.2	18.1	-42.6	18.4	-45.7	18.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.6	-7.8	-5.2	-0.6	4.2	8.9	13.0	15.0	14.4	9.5	4.3	-4.4	-8.8
	DBStd	10.14	8.22	8.47	5.05	3.96	3.45	2.88	2.98	3.29	4.13	5.61	7.29	8.31
	HDD10.0	2816	551	426	329	176	64	7	1	3	58	186	431	584
	HDD18.3	5391	809	659	587	423	293	159	110	127	264	435	681	843
	CDD10.0	479	0	0	0	2	30	98	156	140	44	9	0	0
	CDD18.3	13	0	0	0	0	0	1	7	5	1	0	0	0
	CDH23.3	433	0	0	0	1	27	45	191	141	28	1	0	0
	CDH26.7	75	0	0	0	0	5	3	41	21	5	0	0	0
Wind	WSAvg	2.0	2.2	2.2	1.9	2.2	2.1	2.0	1.8	1.7	1.7	2.1	2.0	2.2
Precipitation	PrecAvg	415	33	19	17	23	32	53	56	53	40	32	29	28
	PrecMax	539	98	62	47	49	97	109	138	130	110	70	62	87
	PrecMin	327	2	1	2	2	9	17	8	17	15	8	5	6
	PrecStd	60	24	15	11	12	18	25	33	29	19	16	15	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.2	9.8	13.6	20.9	27.4	27.5	29.8	28.8	26.6	21.7	8.7	6.5
		MCWB	1.5	3.9	5.1	10.1	13.4	13.5	15.1	15.5	14.2	11.8	4.4	1.7
	2%	DB	3.7	7.2	10.3	17.1	22.8	24.9	27.8	26.8	23.1	17.3	6.6	3.9
		MCWB	1.1	2.6	3.2	7.6	11.0	12.7	14.7	14.7	12.5	9.7	3.6	1.1
	5%	DB	2.6	5.4	8.1	14.7	19.9	22.8	25.5	24.8	20.2	14.6	5.0	2.2
		MCWB	0.6	1.7	2.4	6.4	9.9	12.2	13.7	14.1	11.7	8.2	2.5	0.2
	10%	DB	1.4	3.9	6.2	11.9	17.1	20.6	23.1	22.6	17.2	11.8	3.2	0.8
		MCWB	-0.3	1.1	1.5	5.0	8.5	11.0	13.0	13.2	10.3	6.6	1.2	-0.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	4.8	5.7	10.7	13.6	14.6	16.2	16.5	14.6	12.0	5.1	2.5
		MCDB	4.1	9.1	11.0	19.5	25.9	23.2	26.5	25.2	25.2	21.1	7.9	5.3
	2%	WB	1.5	3.2	4.1	8.1	11.6	13.6	15.2	15.6	13.1	10.1	4.1	1.3
		MCDB	3.0	6.2	9.0	15.9	21.1	22.5	25.3	24.3	21.7	16.9	6.5	3.6
	5%	WB	0.6	2.1	3.0	6.8	10.3	12.7	14.4	14.8	11.9	8.5	2.7	0.3
		MCDB	2.4	4.7	7.0	13.2	18.6	21.0	23.5	23.1	19.0	13.8	4.5	2.3
	10%	WB	-0.3	1.2	1.9	5.6	9.0	11.9	13.5	13.9	10.9	7.1	1.4	-0.8
		MCDB	1.5	3.9	5.9	11.4	16.2	19.0	21.6	21.0	16.6	11.0	3.1	0.7
Mean Daily Temperature Range	5% DB	MDBR	8.0	9.5	10.9	11.9	12.2	12.1	12.7	12.9	12.4	10.2	7.8	7.5
		MCDBR	6.9	9.4	13.2	17.7	18.4	17.3	18.4	17.5	18.6	15.2	7.8	6.3
		MCWBR	5.0	6.0	7.5	9.4	8.7	7.5	7.5	7.4	9.2	8.5	5.2	4.4
	5% WB	MCDBR	6.8	8.8	11.9	15.4	15.9	14.4	16.1	15.9	16.6	14.2	7.5	6.2
		MCWBR	5.2	5.8	7.1	8.5	7.9	6.5	6.9	6.9	8.4	8.0	5.2	4.5
Clear-Sky Solar Irradiance	taub	0.256	0.263	0.298	0.347	0.364	0.388	0.365	0.366	0.336	0.298	0.262	0.251	
	taud	2.320	2.363	2.271	2.155	2.256	2.270	2.386	2.374	2.449	2.507	2.390	2.248	
	Ebn at Noon	749	866	892	882	884	863	878	856	844	815	739	671	
	Edh at Noon	58	76	103	133	127	127	111	106	87	67	54	51	
All-Sky Solar Radiation	RadAvg	0.80	1.66	2.69	4.15	4.96	5.02	5.33	4.58	3.16	1.75	0.84	0.63	
	RadStd	0.08	0.18	0.27	0.39	0.51	0.44	0.47	0.28	0.37	0.20	0.10	0.07	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page